DEPARTMENT OF MATHEMATICS

NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA

B.Tech. I Semester (**EE/CE/ECE/ME/RA/SET**) | **Mid Term – I Examination**

Date: Sept. 24, 2025 | **Max. Marks**: 15 | **Time**: 11:00 am – 11:50 am

Paper: Differential Calculus and Differential Equations (MAIC 101) | **All questions carry equal marks**

Instructions:

- The calculator is not allowed into the examination hall, No extra answer sheets will be provided.

4. **a).** Given $A = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ -1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}$ is an invertible matrix and $\alpha = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \beta = \begin{bmatrix} 1 \\ 0 \\ 2 \\ -1 \end{bmatrix}, \gamma = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \delta = \begin{bmatrix} 1 \\ 0 \\ 2 \\ 0 \end{bmatrix}$ are Linearly Independent over $\mathbb{R}$. Determine whether $A\alpha$, $A\beta$, $A\gamma$ and $A\delta$ are Linearly Independent over $\mathbb{R}$?

   b). Also, determine whether a solution of $AX = \alpha$, can be expressed as the sum of two non-zero solutions, say X_1 and X_2 such that X_1 is the solution corresponding to $AX = 0$ and X_2 is the solution corresponding to $AX = \alpha$?

   c). Is the system $AX = b$ consistent for every choice of b?

5. $Q(x_1, x_2, x_3) = 2x_1^2 + 2x_2^2 + 2x_3^2 - 2x_1x_2 - 2x_2x_3 - 2x_1x_3$ is a quadratic form in three variables over $\mathbb{R}$.

   a). Express $Q(x_1, x_2, x_3)$ in matrix vector form. **b).** Write the type of the matrix A associated with $Q(x_1, x_2, x_3)$.

   c). Find the eigen values of A. **d).** Write the canonical form of $Q(x_1, x_2, x_3)$.

   e). Find the eigen vectors of A. **f).** Find the transformation which transform $Q(x_1, x_2, x_3)$ into its canonical form

   g). Find the orthogonal transformation which transform $Q(x_1, x_2, x_3)$ into its canonical form.

   h). Find the rank, index and signature.

   i). Is the definition of rank, as used in the above question, applicable to every square matrix?

6. Show that $f(x, y) = \begin{cases} \frac{x^2 - y^2}{x - y} : (x, y) \neq (1, -1) \\ 0 : \quad (x, y) = (1, -1) \end{cases}$ is continuous at (1,-1) and $f_x(1, -1)$ and $f_y(1, -1)$ exists.